

One-Channel Grushin Deflation of a Nonnormal Complement Resolvent

Exact Lifted-Inverse Budgets at a Quadratic Band-Merging Map
and a Certified Accretivity Obstruction

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Abstract

An arcwise primal–dual Feshbach enclosure for a stored finite noisy transfer matrix reduced its remaining matrix Rouché condition to an upper bound for the nonnormal complement resolvent $(zI - B)^{-1}$. Direct global smallest-singular-value calculations are expensive and, without validation, provide only lower estimates for the inverse norm. This paper isolates the observed near-singularity before asking for a global certificate.

For $A = z_0I - B$, unit directions obtained by exact normalization of stored vectors u, v , a positive scalar $\widehat{s}$, and residuals

$$r = Av - \widehat{s}u, \quad q = A^*u - \widehat{s}v,$$

we introduce the one-channel lift $\widetilde{A} = A + (\tau - \widehat{s})uv^*$. An exact Sherman–Morrison/Grushin calculation proves that any validated bound $K \geq \|\widetilde{A}^{-1}\|_2$ implies

$$\|A^{-1}\|_2 \leq K + \frac{|\tau - \widehat{s}|(1 + K\|r\|_2)(1 + K\|q\|_2)}{\tau(\widehat{s} - |\tau - \widehat{s}|K\|r\|_2)}.$$

The formula yields a downward conditional budget K_*^- for the lifted inverse and combines with a center-to-arc Neumann bound.

At the budget-tightest RH-28 arc of each of seven stored scales, dimensions range from 2048 to 204800. Componentwise stored-factor arithmetic and Arb-backed exact normalization enclose the right and left residuals by at most 4.38×10^{-10} and 3.56×10^{-9} . Floating inverse iteration gives arc inverse candidates from 4.57×10^2 to 1.76×10^4 . After one-channel lifting, the candidate bulk singular value is 11.1–34.1 times larger than the dangerous value. The conditional lifted-inverse budgets exceed the corresponding floating candidates by factors from 95.0 to 1.73×10^3 ; the finest budget is 6.29×10^4 versus a candidate 6.62×10^2 .

A separate outward-rounded three-vector witness proves that the origin lies in the compressed numerical range of the coarse lifted operator. Hence no rotation can make its Hermitian part positive definite: simple accretivity cannot supply the missing bulk bound even after rank-one lifting.

The algebra, stored residuals, conditional budgets, and numerical-range witness are rigorous for the stated stored finite model. No upper bound for $\|\widetilde{A}^{-1}\|_2$ is proved, so neither the original complement resolvent, a full-contour Rouché transfer, a root count, a continuum limit, nor any statement about the Riemann hypothesis is claimed.

Keywords: Grushin problem; nonnormal resolvent; singular deflation; Sherman–Morrison formula; componentwise enclosure; numerical range; interval arithmetic.

MSC 2020: 47A10; 47A55; 47B65; 65F10; 65F15; 65F35; 65G20; 65P30.

1 Introduction

The packet-complement program considered here studies finite noisy transfer matrices near a quadratic band-merging regime. A packet Feshbach reduction has rank between four and nine, while its ambient complement reaches dimension 204800. The preceding layers constructed a holomorphic shifted Arnoldi model, derived directional and primal-dual residual identities, replaced ordinary residuals by outward-rounded stored-factor enclosures, and extended those enclosures from contour nodes to an exact adaptive cover of a mathematical circle [1–5].

The last step isolated a sharp but difficult condition. On every accepted arc a , the exact stored finite Feshbach map F and its stored rational model $\hat{F}_J$ satisfy a conditional estimate

$$\left\| \hat{F}_J(z)^{-1}(F(z) - \hat{F}_J(z)) \right\|_2 \leq \bar{\eta}_a + M\bar{c}_a, \quad z \in a, \quad (1)$$

where M must be an externally validated upper bound for $\|(zI - B)^{-1}\|_2$. The archived threshold $M_{*,a}^- = (1 - \bar{\eta}_a)/\bar{c}_a$ is admissible but is not an inverse estimate. At the finest stored scale its minimum is only 1.10×10^5 .

There are two reasons not to attack this condition with an unstructured smallest-singular-value calculation. First, a computed singular value and small residual do not exclude an unseen smaller singular value. Second, the complement is strongly nonnormal: a global norm can be governed by a very small number of directions, while the rest of the space is much better conditioned [6, 7]. A useful next step should separate those two effects.

The present paper makes that separation with one rank-one lift. The main contributions are:

1. an exact one-channel inverse bound that converts a certificate for a lifted, better-conditioned operator into a certificate for the original complement shift;
2. a downward conditional budget for the lifted inverse, including exact center-to-arc transport;
3. a seven-scale audit at the budget-tightest RH-28 arc, with componentwise stored-factor residuals and Arb-backed exact normalization;
4. a certified numerical-range obstruction showing that rank-one lifting does not make a simple accretivity proof available.

The outcome is a genuine reduction, not a closure theorem. At the finest selected arc, the original floating margin over the RH-28 threshold is only 6.24. The one-channel reduction asks instead for a bound $\|\hat{A}^{-1}\|_2 < 6.29 \times 10^4$, while its floating candidate is 6.62×10^2 . This roughly 95-fold separation identifies a much more forgiving target for a validated sparse or block-Grushin inverse method.

1.1 Evidence hierarchy

Three levels are kept separate throughout.

1. *Exact stored-model statements*: the rank-one identities, inverse inequalities, normalization bounds, componentwise residual enclosures, downward budgets, and the numerical-range witness.
2. *Conditional statements*: if a validated upper bound for the lifted inverse lies below the tabulated K_*^- , then the original center inverse and its selected RH-28 arc satisfy the required bound.
3. *Floating evidence*: inverse-iteration singular values, candidate inverse norms, gap ratios, and GMRES iteration counts.

No floating candidate is used as an upper bound in a theorem.

2 The selected complement shifts

For the stored packet/complement decomposition, let

$$B = QU^2Q, \quad Q = I - VW, \quad (2)$$

where U is the stored Perron/parity-extracted one-step factor and $WV = I$ on the packet space. The ambient shifted complement is

$$A(z) = zI - B. \quad (3)$$

All matrices are finite. The sparse transfer factor, peripheral modes, packet maps, and spectral parameters are the stored binary64 values used by RH-24–RH-28.

For each noise scale, RH-28 archived an exact dyadic circle partition. We select the arc with the smallest $M_{*,a}^-$, write its disc center and radius as z_0 and ρ , and abbreviate

$$A = A(z_0).$$

This is the tightest *budget* arc. It need not be the point of largest resolvent on the full contour; the experiment therefore does not replace a full-contour audit.

Lemma 2.1 (Center-to-arc transport). *Suppose $A(z_0)$ is invertible and $\|A(z_0)^{-1}\|_2 \leq M_0$ with $\rho M_0 < 1$. Then for every $|z - z_0| \leq \rho$,*

$$\|A(z)^{-1}\|_2 \leq \frac{M_0}{1 - \rho M_0}. \quad (4)$$

Proof. Since $A(z) = A(z_0) + (z - z_0)I$, factor

$$A(z) = A(z_0)(I + (z - z_0)A(z_0)^{-1})$$

and apply the Neumann lemma. □

If L_a is the downward RH-28 threshold and ρ_a is an upper arc-disc radius, it is sufficient to prove the center bound

$$\|A(z_0)^{-1}\|_2 < T_a^- := \text{down} \frac{L_a}{1 + \rho_a L_a}. \quad (5)$$

The implementation evaluates the denominator upward and the quotient downward.

3 One-channel Grushin lift

Let $u, v \in \mathbb{C}^n$ be unit vectors, let $\hat{s} > 0$, and define the two stored-direction residuals

$$r = Av - \hat{s}u, \quad q = A^*u - \hat{s}v. \quad (6)$$

The vectors need not be exact singular vectors. Choose a lift $\tau > 0$ and put

$$c = \tau - \hat{s}, \quad \tilde{A} = A + cuv^*. \quad (7)$$

Then

$$\tilde{A}v = \tau u + r, \quad \tilde{A}^*u = \tau v + q. \quad (8)$$

For an exact singular pair, the lift replaces the singular value $\hat{s}$ by τ and leaves the other singular directions unchanged. For an approximate pair, [equation \(8\)](#) quantifies the failure of that ideal block separation.

Theorem 3.1 (One-channel lifted-inverse bound). *Assume $\tilde{A}$ is invertible and*

$$\|\tilde{A}^{-1}\|_2 \leq K. \quad (9)$$

If

$$\delta_K := \hat{s} - |c|K \|r\|_2 > 0, \quad (10)$$

then A is invertible and

$$\boxed{\|A^{-1}\|_2 \leq \Phi(K) := K + \frac{|c|(1 + K\|r\|_2)(1 + K\|q\|_2)}{\tau(\hat{s} - |c|K\|r\|_2)}}. \quad (11)$$

Proof. The first identity in [equation \(8\)](#) gives

$$\tilde{A}^{-1}u = \frac{1}{\tau}(v - \tilde{A}^{-1}r), \quad \|\tilde{A}^{-1}u\|_2 \leq \frac{1 + K\|r\|_2}{\tau}. \quad (12)$$

The adjoint identity similarly gives

$$\tilde{A}^{-*}v = \frac{1}{\tau}(u - \tilde{A}^{-*}q), \quad \|\tilde{A}^{-*}v\|_2 \leq \frac{1 + K\|q\|_2}{\tau}. \quad (13)$$

Since $A = \tilde{A} - cuv^*$, the Sherman–Morrison denominator is

$$d = 1 - cv^*\tilde{A}^{-1}u.$$

Using [equation \(12\)](#) before taking norms,

$$d = \frac{\hat{s}}{\tau} + \frac{c}{\tau}v^*\tilde{A}^{-1}r, \quad |d| \geq \frac{\hat{s} - |c|K\|r\|_2}{\tau} = \frac{\delta_K}{\tau} > 0.$$

Thus the rank-one inverse formula is valid:

$$A^{-1} = \tilde{A}^{-1} + \frac{c}{d}\tilde{A}^{-1}uv^*\tilde{A}^{-1}.$$

Insert the two solve bounds and the lower bound for $|d|$ to obtain [equation \(11\)](#) [8, 9]. □

Remark 3.2. The theorem does not assume that $\hat{s}$ is the smallest singular value. It is therefore legitimate to treat the computed binary64 vectors and scalar as exact stored inputs. Their quality enters only through $\|r\|$ and $\|q\|$.

For fixed nonnegative residual bounds, $\Phi(K)$ is increasing on the admissible interval $\delta_K > 0$. This produces the next conditional gate.

Corollary 3.3 (Lifted bulk budget). *Let T_a^- be the center budget in [equation \(5\)](#). Define*

$$K_{*,a}^- = \text{down sup}\{K \geq 0 : \delta_K > 0, \Phi(K) < T_a^-\}. \quad (14)$$

If a validated bound satisfies $\|\tilde{A}^{-1}\|_2 < K_{,a}^-$, then the original inverse satisfies the RH-28 requirement throughout the selected arc.*

The code computes [equation \(14\)](#) by monotone bisection with every scalar majorant rounded upward and the final budget rounded downward. In the experiments $\tau = 1$.

4 Exact normalization of stored directions

Inverse iteration produces stored vectors u_0, v_0 whose floating norms are close to one but not exactly one as real numbers. Let

$$\alpha = \|u_0\|_2, \quad \beta = \|v_0\|_2, \quad u = u_0/\alpha, \quad v = v_0/\beta.$$

The exact norms of the binary64 vectors are enclosed by 160-bit Arb sums of squares. No floating normalization is silently promoted to an exact unit identity.

Suppose the raw stored equations have outward residual bounds

$$\|Av_0 - \hat{s}u_0\|_2 \leq R_0, \quad \|A^*u_0 - \hat{s}v_0\|_2 \leq Q_0. \quad (15)$$

Proposition 4.1 (Normalization transfer). *For the exact unit vectors u, v above,*

$$\|Av - \hat{s}u\|_2 \leq \frac{R_0 + \hat{s}|\alpha - \beta|}{\beta}, \quad (16)$$

$$\|A^*u - \hat{s}v\|_2 \leq \frac{Q_0 + \hat{s}|\alpha - \beta|}{\alpha}. \quad (17)$$

Proof. For the first equation,

$$Av - \hat{s}u = \frac{Av_0 - \hat{s}u_0}{\beta} + \hat{s}u_0 \left(\frac{1}{\beta} - \frac{1}{\alpha} \right).$$

Since $\|u_0\|_2 = \alpha$, the second term has norm $\hat{s}|\alpha - \beta|/\beta$. The adjoint equation is identical with α and β interchanged. $\square$

The raw residuals in [equation \(15\)](#) are evaluated by the complete componentwise graph inherited from RH-27. The graph includes sparse transfer actions, peripheral subtraction, packet projection, adjoint actions, and spectral shifts. Each stored binary64 factor is treated as exact input; round-to-nearest operation centers are accompanied by outward componentwise radii using conservative γ_k factors [\[10–12\]](#).

5 Floating reconnaissance and seven-scale audit

5.1 Inverse normal iteration

At each selected center, a deterministic starting vector is repeatedly mapped by

$$v_{j+1} \propto A^{-1}A^{-*}v_j. \quad (18)$$

The two inverse actions are computed by unrestarted outer inverse iteration and restarted GMRES inner solves. This is a reconnaissance tool, not a validated inverse. At the coarse scale, one-channel inverse iteration reproduces a PROPACK candidate to displayed precision while reducing the wall time from about 130 seconds to 1.3 seconds.

The final stored direction is used to define $u_0 = Av_0/\|Av_0\|_2$ and $\hat{s} = \|Av_0\|_2$ in binary64. The stored-factor graph then reevaluates both residuals independently of the inner GMRES stopping tests. Triplet vectors, source hashes, result hashes, and software versions are archived.

5.2 Original tightest-arc gate

[Table 1](#) reports the floating center singular candidate, its Neumann transport over the selected RH-28 arc, the archived RH-28 threshold, and their ratio. These ratios are diagnostics only.

Table 1: Floating diagnostics at the budget-tightest RH-28 arc. The fourth column is the center candidate transported over the archived arc disc. The last column is a candidate margin, not a certified resolvent ratio.

σ	n	$\hat{s}$	arc M_{cand}	$M_{*,a}^-$	margin
10^{-2}	2,048	2.2867×10^{-3}	4.5744×10^2	3.9726×10^{13}	8.68×10^{10}
4×10^{-3}	5,120	1.7241×10^{-3}	5.8988×10^2	5.9371×10^{11}	1.01×10^9
2×10^{-3}	10,240	1.4263×10^{-3}	7.0665×10^2	9.9689×10^{10}	1.41×10^8
10^{-3}	20,480	2.3860×10^{-4}	4.2736×10^3	3.1793×10^9	7.44×10^5
5×10^{-4}	40,960	3.0450×10^{-4}	3.3493×10^3	5.0691×10^8	1.51×10^5
2×10^{-4}	102,400	1.1833×10^{-4}	8.6451×10^3	4.0368×10^7	4.67×10^3
10^{-4}	204,800	5.8108×10^{-5}	1.7625×10^4	1.0999×10^5	6.24

The finest point is informative: the observed resolvent scale lies below the RH-28 threshold, but only by a factor 6.24. A loose normwise validation could easily destroy this margin.

5.3 Deflated bulk gate

Table 2 gives the exact stored-model residual majorant $\max(\bar{r}, \bar{q})$, the downward lifted-inverse budget K_*^- , a floating candidate $K_{\text{cand}} = 1/s_{\text{bulk}}$, their ratio, and the observed singular separation $s_{\text{bulk}}/\hat{s}$.

Table 2: One-channel deflation. Residuals and K_*^- are outward/downward stored-model quantities. The bulk inverse, margin, and gap are floating diagnostics.

σ	$\max(\bar{r}, \bar{q})$	K_*^-	K_{cand}	K_*^-/K_{cand}	gap
10^{-2}	4.416×10^{-11}	9.4974×10^3	1.5930×10^1	596	27.45
4×10^{-3}	6.172×10^{-11}	3.4054×10^4	2.9355×10^1	1.16×10^3	19.76
2×10^{-3}	3.908×10^{-10}	8.8744×10^4	6.2931×10^1	1.41×10^3	11.14
10^{-3}	2.703×10^{-10}	2.1275×10^5	1.2283×10^2	1.73×10^3	34.12
5×10^{-4}	8.014×10^{-10}	1.6525×10^5	1.5383×10^2	1.07×10^3	21.35
2×10^{-4}	2.438×10^{-9}	3.4461×10^5	3.3421×10^2	1.03×10^3	25.29
10^{-4}	3.559×10^{-9}	6.2911×10^4	6.6194×10^2	95.0	26.00

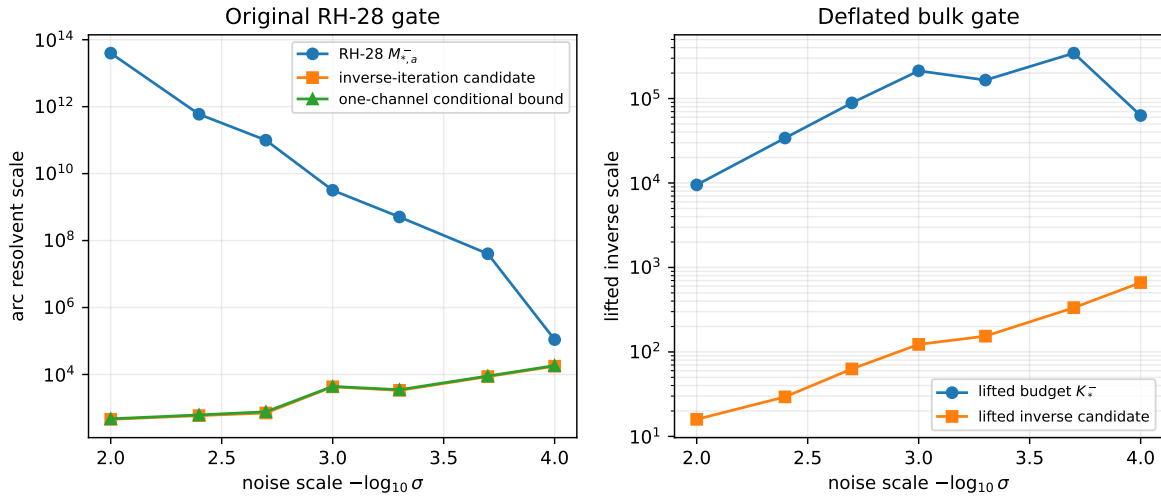


Figure 1: Left: the original RH-28 arc threshold and two floating inverse diagnostics. Right: the exact conditional lifted-inverse budget and the floating lifted candidate. The large vertical separation on the right is the practical gain from one-channel deflation.

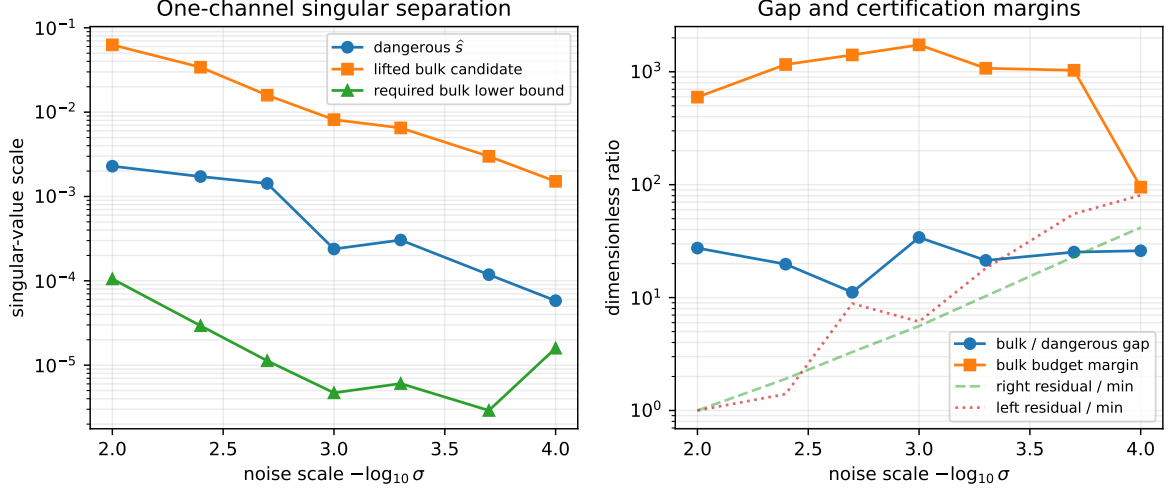


Figure 2: One-channel separation. The lifted bulk candidate exceeds the dangerous singular candidate by factors 11.1–34.1 and lies well above the singular lower bound $1/K_*^-$ required by the conditional theorem.

At the finest selected arc, inserting the floating bulk candidate into the *rigorous formula* Φ gives a conditional center bound 1.80×10^4 and an arc-transported value 1.84×10^4 . This calculation is still floating because K_{cand} has not been proved to upper-bound $\|\tilde{A}^{-1}\|_2$. Its purpose is to show that the exact formula itself does not consume the available RH-28 margin.

6 Why accretivity still fails

A natural way to bound a resolvent is to rotate the operator until its Hermitian part is positive. If

$$\lambda_{\min} \Re(e^{-i\phi} \tilde{A}) \geq \gamma > 0,$$

then $\|\tilde{A}^{-1}\|_2 \leq 1/\gamma$. The one-channel lift removes the smallest observed singular direction, but it cannot change all numerical-range geometry.

Proposition 6.1 (Compressed numerical-range invariance). *For every unit vector x satisfying $v^*x = 0$,*

$$x^* \tilde{A} x = x^* A x. \quad (19)$$

Consequently the numerical range of $\tilde{A}$ contains the convex hull of the compressed quadratic values

$$\{x^* A x : \|x\|_2 = 1, v^*x = 0\}.$$

If that convex hull contains the origin, no rotation of $\tilde{A}$ has a positive-definite Hermitian part.

Proof. The rank-one term contributes $c x^* u v^* x = 0$, proving [equation \(19\)](#). The numerical range is convex by the Toeplitz–Hausdorff theorem [13]. If a rotation had Hermitian part bounded below by $\gamma > 0$, its numerical range would lie in the strict half-plane $\Re(e^{-i\phi} z) \geq \gamma$, which excludes the origin. $\square$

At $\sigma = 10^{-2}$, three stored witness vectors were generated and then reevaluated by the complete componentwise factor graph. Their normalized lifted Rayleigh values are enclosed by

$$\begin{aligned} w_1 &\in [-0.822329596932, -0.822329596907] + i[-0.448800234546, -0.448800234521], \\ w_2 &\in [0.217860304077, 0.217860304097] + i[-1.146076361491, -1.146076361471], \end{aligned}$$

$$w_3 \in [0.011722834533, 0.011722834558] + i[0.221374605040, 0.221374605065].$$

An Arb interval solve for barycentric weights gives

$$\lambda_1 \in [0.048224961594, 0.048224961982],$$

$$\lambda_2 \in [0.138253914403, 0.138253914615],$$

$$\lambda_3 \in [0.813521123614, 0.813521123792],$$

with $\lambda_1 + \lambda_2 + \lambda_3 = 1$ and $\lambda_1 w_1 + \lambda_2 w_2 + \lambda_3 w_3 = 0$. Every weight has a strictly positive lower endpoint. Thus the origin-in-convex-hull statement is a stored-model certificate, not a plot-based inference.

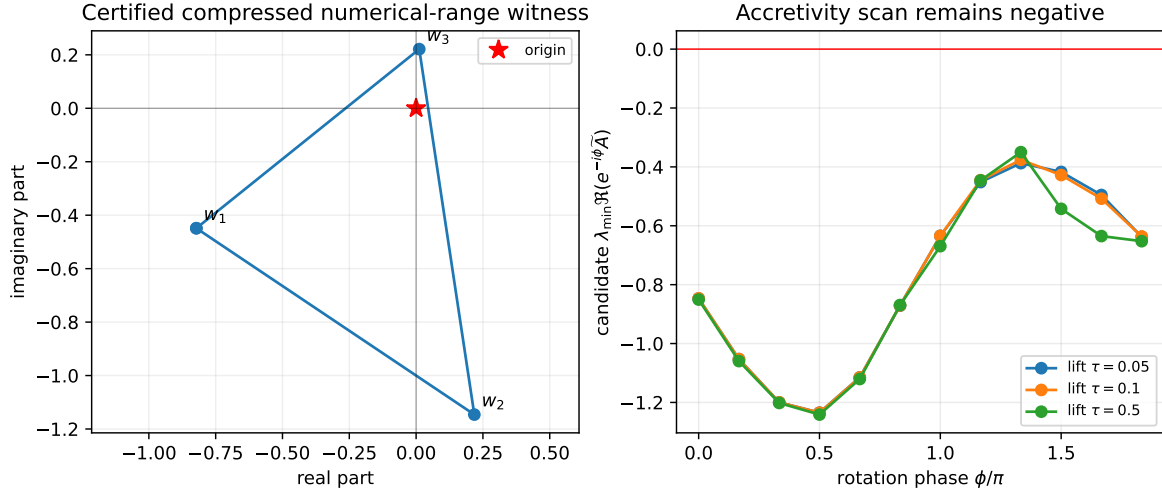


Figure 3: Left: the certified three-point numerical-range witness contains the origin. Interval widths are much smaller than the markers. Right: floating phase scans for three lift values; all sampled minimum Hermitian eigenvalues remain negative, consistent with the exact obstruction.

This no-go result is limited but useful. It does not rule out a validated inverse for $\tilde{A}$. It rules out obtaining that inverse merely by a global rotated-Hermitian coercivity estimate. A future certificate must use more structure: a sparse approximate inverse, a further block decomposition, or a validated normal-equation/Grushin solve.

7 What is proved and what remains

The exact stored-model achievements are:

1. the one-channel residual identities and the inverse bound [theorem 3.1](#);
2. downward center and lifted-inverse budgets for a selected arc;
3. componentwise stored-factor right and left residual enclosures at all seven budget-tightest arcs;
4. Arb-backed exact normalization of every stored dangerous direction;
5. a certified coarse-scale numerical-range obstruction to any global accretivity proof after rank-one lifting.

The following are *not* proved:

1. No upper bound for $\|\tilde{A}^{-1}\|_2$ is supplied. The tabulated K_{cand} values are floating candidates.

2. Only the budget-tightest RH-28 arc at each scale is audited here. A complete contour needs validated lifted bounds on a covering family and a certified transport of the dangerous channel.
3. The RH-28 base winding, interior analyticity, and complement pole count are not validated.
4. The stored sparse transfer matrix is not enclosed relative to an exact Gaussian integral operator, exact critical constants, or exact spectral projectors.
5. No small-noise limit, self-adjoint generator, Hilbert–Pólya construction, Riemann-zero identification, or Riemann-hypothesis consequence is asserted.

The main practical conclusion is quantitative. At the finest selected arc, the original observed resolvent has only a factor-6.24 margin over the RH-28 threshold. After one-channel deflation, the missing validated bulk inverse may be almost two orders of magnitude larger than its observed candidate and still close the exact bound. The dangerous direction is therefore worth handling analytically rather than forcing a global norm estimator to resolve it together with the bulk.

8 Conclusion and next gate

RH-28 reduced the contour comparison to a complement-resolvent upper bound. The present paper does not manufacture that bound from floating singular values. Instead, it proves an exact rank-one reduction that removes the one observed near-singular channel and replaces the original inverse gate by a better-conditioned lifted-bulk gate. Across seven selected arcs, the observed singular separation is at least 11.1, and the conditional lifted budgets exceed the corresponding candidates by at least 95.0.

The next falsifiable task is now narrower:

$$\text{construct a validated upper bound for } \left\| \tilde{A}(z_0)^{-1} \right\|_2 \text{ below } K_{*,a}^-, \quad (20)$$

first at the finest selected center and then on a contour covering family. The numerical-range certificate proves that a one-line accretivity estimate cannot do this. Promising remaining tools are a validated sparse approximate inverse for the lifted operator, a two-level block-Grushin factorization, or an interval normal-equation solve that explicitly retains the lifted channel. Only after that gate is closed should the program return to base winding, interior pole counting, and finite-section/continuous-model questions.

A Outward scalar conventions

Every nonnegative scalar addition and multiplication in the budget code is followed by a successor operation toward $+\infty$. Positive denominators are rounded toward zero before upper division; final admissible budgets are rounded toward zero. Vector norms are evaluated by Arb at 160 bits from the exact binary64 real and imaginary components. The numerical-range witness uses 192-bit Arb interval linear algebra. Componentwise sparse and dense products use conservative operation-count factors under the standard round-to-nearest model, assuming finite intermediates and no harmful underflow.

B Reproduction

From this directory, the unit tests and figures are reproduced with

```
/root/math/.venv/bin/python -m pytest -q
/root/math/.venv/bin/python experiments/make_figures.py
```

One stored scale is regenerated with, for example,

```
OPENBLAS_NUM_THREADS=16 /root/math/.venv/bin/python \  
experiments/run_deflated_certificate.py --sigma 0.0001
```

The expensive calculations are run one physical scale per process. The archived triplet arrays, compact summaries, interval witnesses, figures, source hashes, input hashes, and result hashes are included with the paper.

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